NLP Surprise Test -2

(9/2/24 , 9:00-10:00 a.m.)

(Attempt as much as you can, Total Marks- 35)

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8. Find the shortest string/s not in the language represented by the regular expression $a^*(ab)^*b^*$. (2)
9. For the two regular expressions given below,

   (a) find a string corresponding to r_2 but not to r_1 and (2.5)

   (b) find a string corresponding to both r_1 and r_2. (2.5)

   $r_1 = a^* + b^*$ $\quad r_2 = ab^* + ba^* + b^*a + (a^*b)^*$

10. Find a regular expression corresponding to the language of strings of even lengths over the alphabet of { a, b } (5)
11. Draw a DFA for the language accepting 101(0 + 1)* (5)
12. Draw a DFA which accepts a string containing "ing" at the end of a string in a string of {a-z}, e.g., DFA will accept "anything" but not "anywhere". (5)
13. Draw a DFA which accepts a string containing "the" anywhere in a string of {a-z}, e.g., DFA will accept "there" but not "those". (5)
14. Give context-free grammars that generate the following languages.

    a. $\{ w \in \{0, 1\}^* \mid w \text{ contains at least three 1s} \}$ (4)

    b. $\{ w \in \{0, 1\}^* \mid w = w^R \text{ and } |w| \text{ is even} \}$ (4)

Indian Institute of Technology Bhubaneswar

School of Electrical Sciences

Mid Semester (Spring) Examination – 2024

Subject Name: Natural Language Processing
Date: 23/02/2024
Duration: 2 Hours

Subject Code: CS6L027
Time: 2:30 P.M. - 4:30 P.M.
Full Marks: 60

Instructions

- Write answers as precisely as possible. Stories will negatively affect your marks.
- Make necessary and appropriate assumptions. Write your assumptions clearly before answering a question.
- No notebooks are allowed.

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1. Multiple choice questions [There could be multiple options which are correct, no partial marking] (**10 * 1 mark**)

   1.1. How many word bi-grams can be generated from the given sentence (without removing stopwords)?

   NPTEL videos are a great source to learn engineering courses

   a. 10 b. 9 c. 7 d. 8

   1.2. Machine Translation model-

   a. Converts one human language to another
   b. Converts human language to machine language
   c. Converts one machine language to another
   d. Converts machine language to human language

1.3. Why are low-resource languages challenging for NLP?

a. Because they are always written in non-Latin scripts
b. A type of punctuation mark
c. Due to the limited availability of text data and resources for these languages
d. Because they lack any linguistic structure

1.4. Find a regular expression corresponding to the language of all strings over the alphabet { a, b } that do not end with ab

a. (a+b) (a+bb)* b. (ab)*(aa+bb) c. (a+b)*(a+bb) d. (a+b)(a+bb)

1.5. Which is/are the example/s of Under-stemming

a. adhere → adher
b. adhesion → adhes
c. appendicitis → append
d. append → append

1.6. Identify the word which has different sound represented by two underlined letters (ch) among the following four words

a. Church b. Epoch c. Each d. Reach

1.7. 'I never said she **stole** my money' - The stress is on the **bold** word in this sentence. What does the sentence intend to mean

a. I simply didn't ever say it.
b. I might have implied it in some way, but I never explicitly said it.
c. I said she stole someone else's money.
d. I just said she probably borrowed it.

1.8. In inverted index term-document representation, the postings list should be sorted by

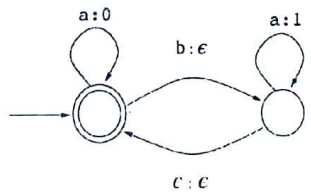
a. Document Frequency b. Term Frequency c. Term ID d. Doc ID

1.9. Which of the following tasks can use TF-IDF vector representation

a. Document classification b. Information Retrieval

c. Stop word filtering d. Topic Modeling

1.10. What output/s will this FST produce, given the input **abaaab**?



a. 0111 b. 0000 c. 0101 d. 0100

2.

2.1. As defined in the following table, are X and Y independently distributed? (2)

x	0	0	1	1
Y	a	b	a	b
p(X = x, Y = y)	0.3	0.1	0.2	0.4

2.2. Compute the entropies for:

a) H(X),H(Y) (1+1)

b) H(X, Y), H(X|Y), H(Y |X) (1.5+1.5+1.5)

c) D(X||Y) (1.5)

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3. Given a grammar G, defined by:

S → NP VP

NP → N | D N | NP PP

VP → V NP | V NP PP

PP → P NP

N → I | elephant | pajamas | man | telescope

V → shot | lost | held | saw

P → in | with

D → my | an | a | the

3.1. Generate 2 grammatically correct sentences with at least 10 words using this grammar. Show the parse trees. (1.5+1.5)

3.2. Give all possible derivations of the following sentence. Show the parse trees. (2)

I saw the man with the telescope

3.3. Do the following sentences belong to L(G)? (1.5+1.5)

3.3.1. **the I lost my I**

3.3.2. **the elephant lost all pajamas**

3.4. Alter the grammar G to grammar H to allow for modifying adjectives by adding

A → bold | green | large

What other rules have to be modified to accept sentences like: (2)

the green elephant saw a large telescope

Provide the transition functions of H.

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4. Consider the following training text:

SALAMI IST IM SALAT. LAMA IST ĂLT.

4.1. Create an MLE n-gram model for **letter unigrams**. (Note: Consider both space and period as separate letters.) (2)

4.2. Create an MLE n-gram model for **letter bigrams** and draw the corresponding Markov Chain. Assume a start symbol. (2)

Consider the following test text: LAMA ISST

4.3. Compute the probabilities of the test text for each model. (2+2)

4.4. Compute the perplexities of the models on the test text. (2+2)

4.5. Compute the probabilities and perplexities using a mixture model with $\lambda_1 = 0.1$ and $\lambda_2 = 0.9$ on the MLE estimates. What can you say about the quality of the model? (3)

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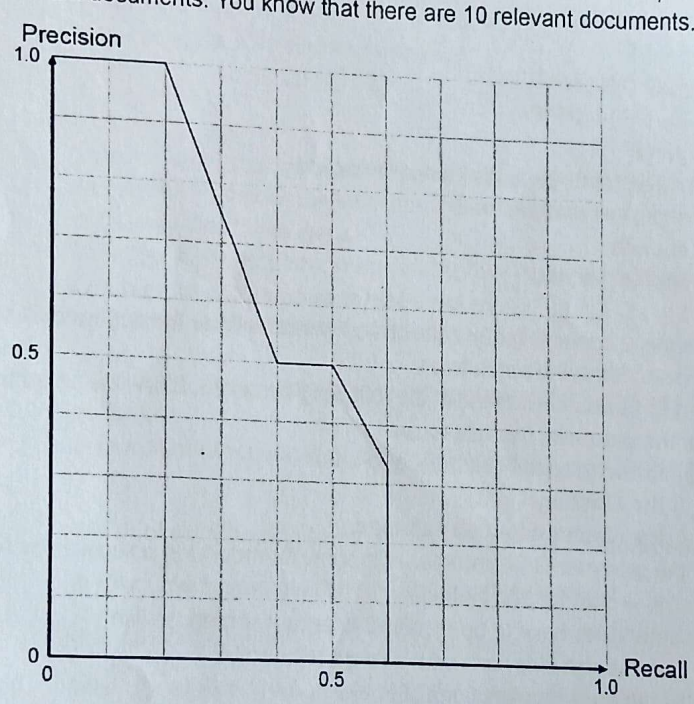
5. Assume a document retrieval system produced the following interpolated ROC curve (precision-recall curve) on a particular query (based on 20 hits). Hits signify retrieving relevant documents. You know that there are 10 relevant documents.



5.1. What is the precision after the system has retrieved 3 relevant documents? (2)

5.2. Going down the hit list, you discover that you have retrieved ***n*** documents, and all of them are relevant. What's the maximum possible value of n? (1)

5.3. Where are the relevant documents in the hit list (ranked)? Mark a relevant document with an R in the corresponding box. Leave irrelevant documents Unmarked. (2)

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20

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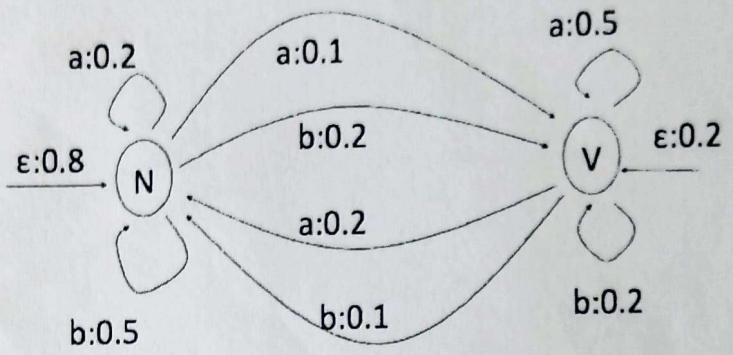
6. Consider the HMM given in the figure:



6.1. Compute the probability **P(bbba)** of the observation using the **forward algorithm.** (5)

6.2. What is the sequence of states that marks the path with the highest probability for **abaab**, and what is the probability?(Hints: **Viterbi Algorithm**) (5)

Indian Institute of Technology Bhubaneswar

School of Electrical Sciences

End Semester (Spring) Examination – 2024

Subject Name: Natural Language Processing
Date: 29/04/2024
Duration: 3 Hours

Subject Code: CS6L027
Time: 3:00 P.M. - 6:00 P.M.
Full Marks: 80

Instructions

- Write answers as precisely as possible. Stories will negatively affect your marks.
- Make necessary and appropriate assumptions. Write your assumptions clearly before answering a question.
- No notebooks are allowed.

1. Multiple choice questions **(5 * 1 mark)**

1.1. To compute the likelihood of a sentence using a bigram model, you would:

a. Calculate the conditional probability of each word given all preceding words in a sentence and multiply the resulting numbers
b. Calculate the conditional probability of each word given all preceding words in a sentence and add the resulting numbers
c. Calculate the conditional probability of each word in the sentence given the preceding word and multiply the resulting numbers
d. Calculate the conditional probability of each word in the sentence given the preceding word and add the resulting numbers

1.2. In a dependency parse

a. All nodes are labelled with words
b. Only leaf nodes are labeled with words
c. Only non-terminal nodes are labelled with words
d. No words appear in the tree

1.3. N-fold cross-validation is a technique for evaluation that uses

a. All data in the corpus for training
b. 10% of the data in the corpus for training
c. 90% of the data in the corpus for training
d. 50% of the data in the corpus for training

1.4. In an HMM, tag transition probabilities measure

a. The likelihood of a POS tag given a word
b. The likelihood of a POS tag given the preceding tag
c. The likelihood of a word given a POS tag
d. The likelihood of a POS tag, given all preceding tags

1.5. In an Early parser, a table is filled. Which of the following does *not* appear as a table entry?

a. Completed constituents and their location
b. Predicted constituents
c. Word predictions
d. In-progress constituents

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2. Short answer type questions. Contain your answer within a maximum of 4-5 sentences. **(15)**

2.1. List all the morphemes in the word "encountered". Which ones are free, and which ones are bound? **(1+2)**

2.2. State the difference between homonymy and polysemy and give an example of each. **(2+1)**

2.3. Define the "word sense disambiguation" task and explain how part-of-speech tagging can be used for word sense disambiguation. **(1+2)**

2.4. Consider a classification model that separates email into two categories: "spam" or "not spam." If you raise the classification threshold (e.g. some probability score), what will happen to the precision of the model? What will happen to recall? **(1.5+1.5)**

2.5. Describe the concept of negative sampling. Why is it a good idea? **(1+2)**

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3. Using the following trigram language model, identify the most likely missing word in the sequence - **"as she said this she ____ to the mountains"**. Show your calculations clearly. **(6)**

P("as she climbed") = 0.03	P("she said yes") = 0.16
P("as she is") = 0.07	P("she talked to") = 0.30
P("as she said") = 0.10	P("she walked to") = 0.35
P("looked to the") = 0.10	P("talked to the") = 0.05
P("not the correct") = 0.04	P("this she did") = 0.01
P("not the mountains") = 0.02	P("this she like") = 0.40
P("said to the") = 0.05	P("this she looked") = 0.30
P("she looked to") = 0.10	P("this she talked") = 0.10
P("she said no") = 0.35	P("this she walked") = 0.10
P("she said this") = 0.05	P("to the mountains") = 0.10
P("she said to") = 0.13	P("walked to the") = 0.10

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4. Calculate the P@5, P@10, and MAP (Mean Average Precision) scores for the two engines shown below and indicate which engine is better. (A ✓ denotes a positive result, and a ✗ denotes a negative result.) Show your calculations clearly. **(2+2+4+1=9)**

Rank	Engine A	Engine B
1	✗	✓
2	✗	✓
3	✗	✗
4	✓	✗
5	✓	✗
6	✓	✓
7	✗	✗
8	✓	✓
9	✗	✗
10	✓	✓
11	✗	✗
12	✗	✓

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5. Given the following grammar and lexicon rules, parse the sentence "**time flies like a fruit like the flies**" with the probabilistic CYK algorithm.

Grammar:

S	$\rightarrow$	$NP\ VP$	1.0
NP	$\rightarrow$	$D\ N$	0.5
NP	$\rightarrow$	$N\ N$	0.3
NP	$\rightarrow$	$NP\ PP$	0.2
VP	$\rightarrow$	$V\ NP$	0.6
VP	$\rightarrow$	$V\ PP$	0.4
PP	$\rightarrow$	$P\ NP$	1.0

Lexicon:

N	$\rightarrow$	*fruit*	0.2
V	$\rightarrow$	*flies*	0.1
N	$\rightarrow$	*flies*	0.1
V	$\rightarrow$	*like*	0.3
P	$\rightarrow$	*like*	0.2
D	$\rightarrow$	*a*	0.4
D	$\rightarrow$	*the*	0.5
N	$\rightarrow$	*banana*	0.1
N	$\rightarrow$	*time*	0.1

5.1. Show the process clearly. **(8)**
5.2. Give the probability of the most likely parse. **(5)**
5.3. Give the probability of the sentence given the grammar. **(2)**

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6. Consider the following POS-Tagged corpus having three sentences is given

S1: **the/D cat/N can/VA fish/VV a/D trout/N in/P seconds/N**
S2: **the/D fish/N swim/VV in/P the/D can/N**
S3: **workers/N can/VV the/D food/N**

6.1. Train a POS bigram Hidden Markov Model from the given corpus using Maximum Likelihood Estimation. (**Hints**: State transition diagram with word emissions, probabilities assigned to the edges) **(8)**

6.2. Tag the following text with the trained HMM and provide the probabilities in each step: **(5)**

the workers can can the fish in a can

6.3. What is the probability of the following text? **(2)**

a cat can

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7. Consider the following sentences are there in Document *D*:

S1: **bears eat cats in the dark**
S2: **when it is dark bears eat cats and dogs**
S3: **dogs eat cats**
S4: **dark bears eat only big cats**

7.1. Compute the TF*IDF feature vectors for all these sentences, only using open class words (no stopwords). Stopword list: in, the, is, it, and, only, when **(8)**

7.2. Compute the Sentence similarity graph by calculating the cosine similarity of their vectors. Apply a threshold of **0.1** . **(2)**

7.3. Apply PageRank (**5 Rounds**) and generate an extractive summary of Document *D* consisting of 2 sentences. Use **0.85** as the damping factor. Initialize page rank uniformly. **(5)**